

Lunar Water Logistics

3.3 Million Tonnes of Lunar Water to Lunar Orbit, on a Fleet That Imports Nothing

Architecture Reference Document | Rev B — September 2026 | Not for Hardware Commitment

1. Executive Summary

Aegis Station is a rotating habitat in a 100 km lunar orbit, shielded by three metres of water around the outer hull of each of its three rings. At the 1g baseline that shield is 3.3 million tonnes. Lifting that mass from Earth is not a plan; it is a number that ends the program. Every tonne of shield water is therefore mined at the lunar south pole, carried to the pad in a standard 45-tonne cartridge, and flown to orbit by an autonomous tanker.

Rev B makes one structural change to the chain. The tanker flies on liquid oxygen and liquid hydrogen split from lunar water at the pad, from its first flight. The earlier revision baselined methalox with methane shipped from Earth and treated hydrolox as a later upgrade. At fleet cadence that methane was a recurring import of hundreds of tonnes a day landed on the Moon, a supply chain larger than every lunar landing to date, repeated indefinitely. Anything a program has to keep buying from Earth is the thing that ends it. The import is designed out before the first flight, and the honest cost moves from the launch manifest to a refinery-class power plant on the surface.

The rest of the architecture is unchanged and is stated here as a baseline, not a closed design: 45 tankers, 2,025 tonnes a day, a fill of roughly 4.5 years, and the Lunar Orbital Propellant Depot as the fleet's first customer before the station exists. Section 10 lists what does not yet close.

2. Water Requirement

The shield is a 3 m water layer on the outer hull of each ring, outboard of a 7 m utility band and a 40 m habitable depth. The requirement is set by geometry and the 1g rotation baseline, not by logistics.

Parameter	Value
Shield thickness	3 m of water, outer hull, all three rings
Shield radial span	47 m to 50 m from the ring centreline, outboard of the utility band
Shield volume, three rings	~3.3 million m ³
Water mass	~3.3 million tonnes
Olympic pool equivalent	~1,320
Source	Lunar south pole, permanently shadowed craters. Earth-sourced water is excluded in every phase.

Shield thickness is the multiplier on every number in this document: tankers, sorties, power, and years. The 3 m figure is held by the station design; the dose case for it is stated on the water-shield evaluation page, not here.

3. Delivery Architecture

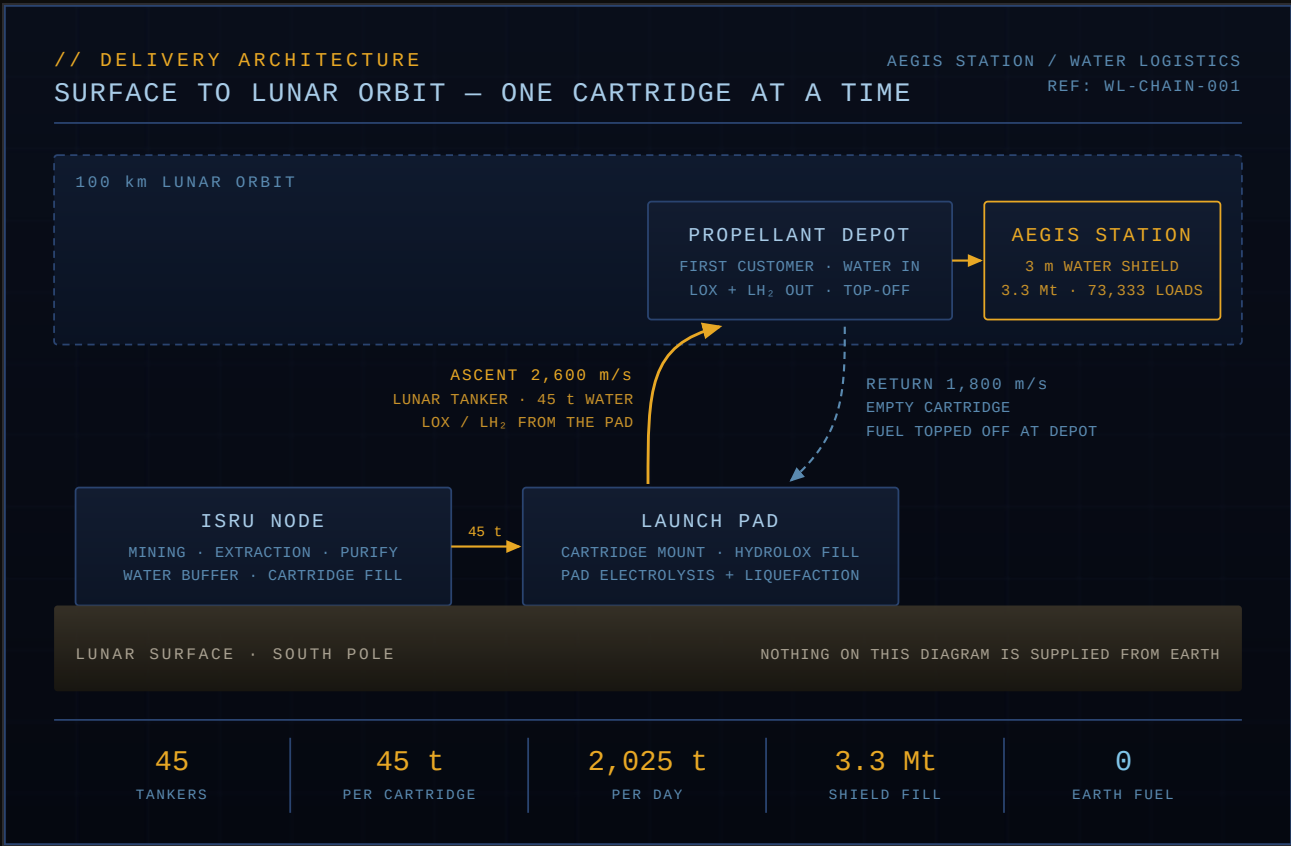


Fig. 1 — The delivery loop. Every sortie carries 45 t of water up and an empty cartridge down. Nothing on the loop is supplied from Earth.

Parameter	Value
Fleet	45 autonomous reusable tankers, VTOL, one ground console
Payload	45 t of water per sortie, in one Modular Tank Cartridge
Ascent	2,600 m/s to 100 km lunar orbit
Return	1,800 m/s, empty cartridge, fuel topped off at the depot
Daily throughput	~2,025 t/day
Fill time	~4.5 years at that throughput
Sorties, total	~73,333
Propellant	LOX / LH ₂ , both split from lunar water at the pad
Earth-supplied consumables	None, in any phase

The Modular Tank Cartridge

The cartridge is the atomic unit of the chain: a 10 m × 2.5 m structural cylinder holding a flexible bladder of ~49 m³, filled by vacuum draw at the ISRU node and emptied by inert-gas displacement at the destination. It is cargo the tanker carries, not part of the tanker, and it has one water port and one gas port that are identical at the node, the pad, the depot, and the station. Water is never a cryogen anywhere in the chain. It needs no insulation, no vacuum-jacketed lines, and no boil-off management, which is why it is the commodity that moves.

Mission profile

- 0 **Depot first light.** A single tanker at low cadence lifts the first cartridges to the Lunar Orbital Propellant Depot and commissions end-to-end propellant production. Full fleet deployment is not required.
- 1 **Shield-fill campaign.** Sustained surface-to-orbit delivery to the station's shield reservoirs. This is the requirement that sizes the 45-vehicle fleet.
- 2 **Depot resupply.** Runs in parallel from first light and continues indefinitely after the fill. The depot is the fleet's first customer and a permanent one.
- 3 **Surface distribution.** Ballistic hops between ISRU and LUNET nodes at 160–940 m/s, redistributing water and LOX. Up to six hops per orbital propellant load.
- 4 **Rapid response.** One or two tankers held in standby; emergency water to a crewed outpost within 6–8 hours of dispatch.

4. Propellant: Hydrolox From Day One

The Moon has water ice. Electrolysis splits it into liquid oxygen and liquid hydrogen, the two highest-performing chemical propellants there are, and the only ones the Moon can make from a single feedstock without importing anything. The tanker burns both from its first flight. There is no methane

phase, no transition, and no import to taper off later. The Short Hopper already flies LOX/LH₂ for the same reason; with this revision every vehicle on the surface burns the same ISRU product.

Parameter	Rev A (methalox)	Rev B (hydrolox)
Propellant	LOX / CH ₄ , methane imported from Earth	LOX / LH ₂ , both from lunar water
Specific impulse, vacuum	~360 s	~450 s
Launch mass	~102 t	~95 t
Ascent propellant	~52 t	~42 t
Round-trip propellant	~57 t	~45 t
Thrust, engines	~165 kN, 3 × 60 kN	~155 kN, 4 × 40 kN, BE-7 class, engine-out capable
Recurring Earth import per sortie	~12.5 t methane	None

Rev B propellant masses follow the rocket equation at the same payload, delta-v, and implied dry mass as Rev A. They are not a new sizing pass; see Section 10.

The hydrogen objection

Liquid hydrogen at 20 K is hard to store and harder to move. This architecture answers that the same way LUNET does: *produce near, consume fast*. A tanker fills from a liquefier a few metres from the pad and is empty in orbit within hours. That is the shortest hydrogen dwell any lunar vehicle will have. The long-dwell problems, boil-off, thermal-cycle fatigue on transfer lines, and embrittlement, belong to depots holding LH₂ for months, not to a vehicle that burns its load the same day it is filled. Zero boil-off storage at 20 K has been demonstrated at twice the previous state of the art, and the industry's own next-generation lunar lander is hydrolox for precisely this reason: it is the propellant ISRU can actually make.

The honest cost

Hydrogen is a ninth of water by mass, so a full tanker load means splitting several times its own hydrogen mass in water and liquefying the result. The tanker pad is therefore not a LUNET fill node, which is sized at tens to a few hundred kilowatts. It is a refinery-class facility with power to match, and it is the same plant the depot and the surface network already assume exists. That is one thing to build, once, on the Moon, rather than something to keep launching. Its size scales with sortie cadence and is an open item in Section 10.

Design rule. No vehicle in the surface fleet may depend on a consumable that must be resupplied from Earth. *Methane failed this rule. Hydrolox passes it because the feedstock is the cargo.*

5. The Depot as First Customer

Every lunar-water architecture faces the same deadlock: surface extraction is hard to justify without an orbital customer, and orbital hardware is hard to justify without a proven resource. The Lunar Orbital Propellant Depot breaks it by being the earliest useful thing in lunar orbit. Its hardware can be emplaced and stand dormant; its commissioning date is set by the first cartridge to arrive, and its existence is what makes that date matter.

- **Water in, propellant out.** Depot electrolysis produces LOX and LH₂ from cartridge water. With a hydrolox fleet the hydrogen is a product, not a vented byproduct: tankers top off both propellants at the depot between cycles, so the loop closes at both ends of the ascent.
 - **Methane for the Hauler only.** The Earth–Aegis Long-Hauler still burns methalox for its return; it brings its own methane as manifest cargo. That import is declared, bounded, and never touches the tanker fleet.
 - **Standard interfaces.** The cartridge and its two ports are the same at the depot as at the pad. A depot that only its builder can dock with is a fuel tank, not infrastructure.
 - **Beachhead.** The depot's pressurised node is the program's first shirt-sleeve presence in lunar orbit, before any ring is assembled.
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6. ISRU and Extraction

Ice-bearing regolith is excavated from permanently shadowed craters near the south pole and heated by microwave or radiant thermal rigs; the ice sublimates, is cold-trapped, condensed, and filtered to a single water grade that serves both cargo and propellant. An ambient surface buffer absorbs production swings and feeds two consumers: cartridge fill for the tanker, and pad-side electrolysis and liquefaction for the tanker's own propellant.

- Deposit assessment precedes extraction; the subsurface ice assessment capability and the WOK prospecting rover are the instruments aimed at that gate.
- Mining, extraction, and transfer are autonomous and modular; power is polar solar with fission for the lunar night.
- For every tonne delivered to orbit, roughly another tonne and a half is split for propellant. The mining rate, not the delivery rate, sizes the node.

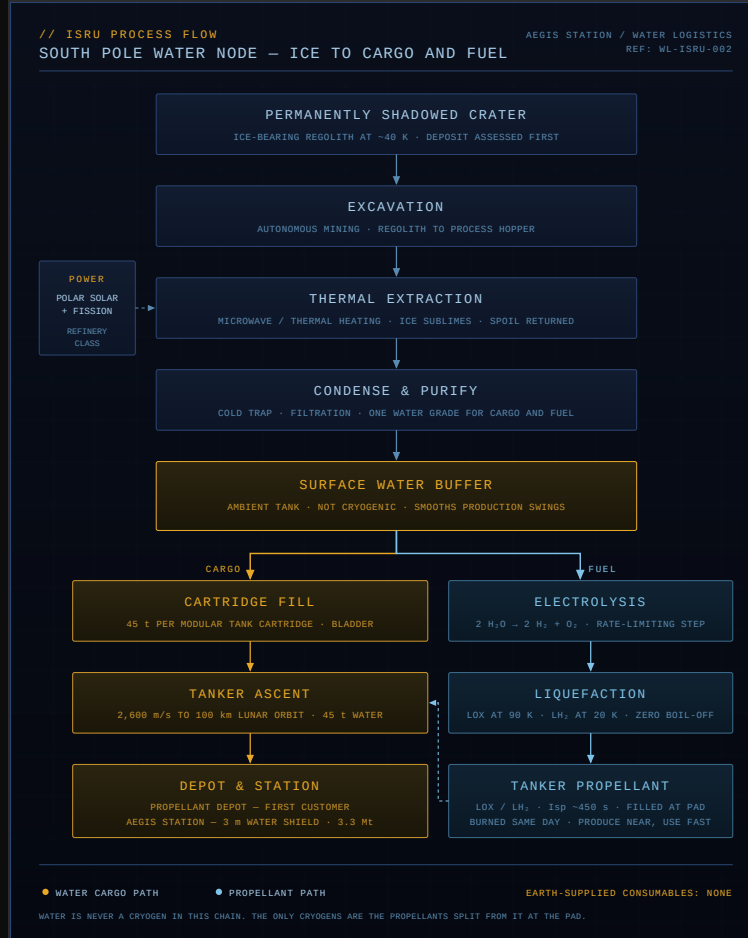


Fig. 2 — From ice in the crater to water in a cartridge and hydrolox in the tanker. The buffer is ambient; the only cryogens are the propellants split from it.

7. Logistics Infrastructure

Surface

- Launch pads with dust control and blast deflection
- Cartridge fill, staging, and ground inventory of displaced inert gas
- Pad-side electrolysis and liquefaction: the refinery that fuels the fleet
- LUNET utility nodes for surface mobility, refuelling, and inter-node exchange
- Fleet Commander ground console: dispatch, pad deconfliction, berth queuing, standby reserve

Orbital

- Lunar Orbital Propellant Depot: cartridge reception, electrolysis, LOX and LH₂ storage, tanker top-off
- Station cartridge docking ports and water transfer lines to the shield reservoirs
- Autonomous transfer monitoring, inventory reconciliation, and fault management

Shield fill

Rings receive shield water from the start of assembly through evenly spaced fill ports. Fill sequencing preserves mass balance across each ring, and spin-up to 1g follows structural and systems verification of the complete ring assembly, so the shield is filled and balanced before the ring rotates.

8. Cost Considerations

Source	Cost per kg delivered	Cost per Olympic pool
Earth-launched	\$2,500	~\$6.25 billion
Lunar-sourced, target	\$150	~\$375 million

Scenario	Total
Earth-launched water, 3.3 Mt	~\$4.1 trillion
Lunar water, delivery only, at \$150/kg	~\$495 billion
Surface refinery, power, mining, and depot	Not yet costed; stated on the program pages as they close

The \$150/kg figure is a program target, not a bottom-up estimate. The earlier revision stated \$247.5 billion for delivery, which is inconsistent with its own per-kilogram figure; Rev B corrects it. Under Rev A the methane import alone, at any plausible landed price, would have exceeded this delivery total every few years of the campaign.

9. Long-Term Applications and Strategic Value

The extraction and delivery chain does not retire when the shield is full. The same fleet, pad, and depot serve depot resupply, surface distribution, emergency response, and propellant staging for anything leaving cislunar space, still without an Earth-supplied consumable.

- LOX and LH₂ production for cislunar and deep-space vehicles
- Depot refuelling in lunar orbit and at Earth–Moon staging points
- Life support reserves and shielding water for future habitats and platforms
- Water as a priced, deliverable orbital commodity with published interfaces
- A replicable model for any station built where water is available

The water economy doesn't end with Aegis. It begins with it.

10. Open Items

This revision fixes the propellant decision and leaves the campaign numbers as stated. The following do not yet close and are carried as open, not hidden.

Scope and assumptions for the fleet analysis

The tanker fleet is analysed as a vehicle cycle between a surface pad and a station berth. Three things are assumed rather than analysed here, because each belongs to another part of the program:

- A1 **A full cartridge is available at the pad whenever a tanker is ready to lift.** Mining, extraction, cartridge fill, and the size of the cartridge pool are upstream of the fleet and sized separately.
- A2 **An empty cartridge is available at the station whenever a tanker berths.** Shield-side discharge and cartridge handling are station systems.
- A3 **Refuelling is timely wherever it happens, surface or station.** Refinery and depot capacity are their own items; they set the ceiling on cadence but are not part of the vehicle cycle.

Under those assumptions the fleet's own limits are pad-hours, berth-hours, orbital phasing, airframe availability, and turnaround mechanics. Where refuelling happens and how propellant gets aboard are both still open and are carried below.

Item	Status
Daily turnaround	2,025 t/day at 45 t per sortie is one flight per tanker per day for the whole campaign, about 1,600 cycles per airframe. That is an airliner cadence in an environment gentler than any reusable booster faces: no reentry, no aerodynamic load, one-sixth-g landings. The airframe can plausibly do it. No reusable orbital-class vehicle has yet flown daily. If turnaround settles at three to five days the fill runs 13 to 22 years with the same fleet.
Engine pipeline	Each engine position sees roughly 500 hours of burn over an airframe life, many times any demonstrated engine life, so engines are scheduled replacement items. At 50–100 sorties per engine set, four engines per vehicle, the campaign consumes about 3,000–6,000 engines, two to four a day from a single printed part number, 450–900 t landed on the Moon. As serial printed 40 kN-class engines at \$0.5–2 M each that is \$3–10 B, one to two percent of the delivery target; at legacy bespoke hydrolox pricing it would be \$60–120 B. Engine life is the lever: 200 sorties per set quarters the count. Life-expired engines stay at the pad as the first refurbishment demand on the surface.
Vehicle dry mass	The launch, payload, and propellant figures imply roughly 5 t of structure, tanks, engines, and cartridge shell. A realistic lander dry fraction would reduce payload or raise launch mass. Not resized in Rev B.
Surface power	Electrolysis and hydrogen liquefaction at fleet cadence are of order hundreds of megawatts continuous. LUNET node sizing does not cover this; the pad refinery is a separate facility and is not yet sized.
Cartridge reuse	The cartridge is infrastructure, not a consumable, but its cycle life and bladder material are undefined. Fleet size of cartridges bounds throughput.
Loading-site capacity	Forty-five sorties a day is one launch every 32 minutes. With six pads and six berths, each tanker may hold a pad for at most 3.2 hours and a berth for at most 3.2 hours, all steps included. The fleet dispatch ADR cites a 4–8 hour nominal swap; at that range six pads cap the fleet at 18–36 sorties a day and the fill runs 6–11 years however many tankers exist. Pads are cheap against megawatts: ten pads relax the turnaround to five hours. A tanker waiting on propellant or a cartridge must not hold a pad; it waits on an apron, so pad occupancy measures servicing only.
Refuelling method and location	Open. Two methods: swappable propellant modules, pre-chilled and filled off the critical path, so pad time is mount time and the same pattern the water cartridge already uses; or fluid connections, a lighter vehicle but chilldown and transfer on every cycle and cryogenic coupling wear at the pad. Two locations: surface only, or surface plus depot top-off for the return leg, which the depot already provides for. The choice sets the turnaround time in the row above and is the next fleet-level trade.

Item	Status
Airframe availability	Engine swaps and inspections take each airframe out of service for a fraction of the campaign. At ~90 % availability, 45 airframes deliver ~40 sorties a day; the 45-a-day figure needs about 50 airframes or accepts the shortfall.
Resource base	Shield water plus propellant water is several million tonnes against polar ice estimates that span two orders of magnitude. Deposit assessment is the gate, and it is why the program is waiting.

11. Changes From the Previous Revision

Item	Previous	Rev B
Tanker propellant	LOX/CH ₄ , methane imported from Earth, hydrolox as a later phase	LOX/LH ₂ from lunar water, first flight onward
Earth-supplied consumables	~12.5 t methane per sortie for the first years	None
Water handling	"Cryogenic transport tanks", "cryogenic storage tanks", "cryogenic transfer lines"	Ambient water in bladder cartridges; only the propellants are cryogenic
Surface nodes	Remote Operations Nodes	LUNET utility nodes plus a refinery-class pad
Depot	Not described	First customer; commissions on first cartridge; hydrogen retained for tanker top-off
Delivery cost	~\$247.5 billion	~\$495 billion, matching \$150/kg × 3.3 Mt
Infrastructure cost	~\$400–500 billion	Not yet costed
Open items	Not stated	Section 10